

quickly than people. As robots become more capable of doing more activities and making more decisions on their own, computing will increasingly be required to take place at the edge of the network. Increased productivity can be found near the edge.

For example, a ‘self-aware’ robot that is powered by data received and processed at the edge can detect the likelihood of its own imminent failure by gathering and processing data at the edge itself. With the help of other robots on the assembly line, the at-risk machine can automatically shut itself down while other robots adjust their workflow in real-time to compensate for the loss of one of their own “workers.” The production line slows down, but it does not come to a halt. A human collaborator can make the necessary repairs, and the system will be back up and running in no time.



According to one definition, there is an “edge-driven paradox”: as robots get increasingly complex, skilled, and dexterous, the amount of effort necessary to educate them decreases in many circumstances. Leading manufacturers recognise that lowering the learning curve for end-users is a critical component of increasing the appeal of industrial robots to a wider range of people.